Reading seminar 7.8
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6.3 Affine Geometric Invariant Theory (GIT) and good moduli spaces (GMS)

§1. Good Moduli Spaces

- **Def'n**: A q-cpt q-sep morphism $\pi:\mathcal{X}\to X$ from an alg. stack to an alg. space X is a GMS if
  1) $\mathcal{O}_X \xrightarrow{\sim} \pi_*\mathcal{O}_{\mathcal{X}}$
  2) $\pi_*: QCoh(\mathcal{X})\to QCoh(X)$ is exact. "good"
- Eg: $[\mathbb{A}^1/\mathbb{G}_m]\to \operatorname{Spec} k$ GMS

Non-Eg: $\boxed{[\mathbb{P}^1/\mathbb{G}_m]}\to\operatorname{Spec} k$ is not a GMS

- 2) fails "$\mathcal{O}(-2)\hookrightarrow\mathcal{O}(-1)\twoheadrightarrow\mathcal{O}_0$"

⚠ 2 closed points upstairs ⇝ 1 pt

Model: Affine GIT

- $G \curvearrowright \operatorname{Spec}A/k$ G linearly reductive $Rep(G)\to Vect_k$ exact. $V\mapsto V^G$

$$\begin{tikzcd}
\operatorname{Spec}A \arrow[d] \arrow[rd, bend left, "\hat{\pi}"] & \\
{[\operatorname{Spec}A/G]} \arrow[r, "\pi"] & \operatorname{Spec}A^G
\end{tikzcd}$$

1) $\Gamma([\operatorname{Spec}A/G],\mathcal{O}_{[\operatorname{Spec}A/G]})=A^G$
2) $\pi_*: QCoh([\operatorname{Spec}A/G])\to QCoh(\operatorname{Spec}A^G)$ is exact.

[Mumford. Seshadri] good quotient/GIT quot

$\operatorname{Spec}A^G = \operatorname{Spec}A/\!/G$

- $M^{ss}(r,d)$
- $M^{K-ss}(d,v)$
- does not allow a GMS

- **Thm 6.3.5**: $\pi:\mathcal{X}\to X$ GMS $\mathcal{X}$ q-sep / S scheme.
  1) π is surj. & universally closed,
  2) $\mathcal{Z}_1, \mathcal{Z}_2 \overset{closed}{\subset} \mathcal{X}$ $\operatorname{im}(\mathcal{Z}_1\cap\mathcal{Z}_2)=\operatorname{im}(\mathcal{Z}_1)\cap\operatorname{im}(\mathcal{Z}_2)$

     For geometric pts $x_1,x_2\in\mathcal{X}(k)$ $\pi(x_1)=\pi(x_2)\in X(k)$ iff $\overline{\{x_1\}}\cap\overline{\{x_2\}}\neq\emptyset$. In particular π induces a bij. between closed pts
  3) $\mathcal{X}$ Noeth. so is X. If $\mathcal{X}$ is of f.t./S and S is Noeth, then X is of f.t./S and π_* preserves coherence
  4) If $\mathcal{X}$ is Noetherian, then π is initial for maps to alg. spaces.
- **Cor 6.3.7**: G lin. reductive gp / $k=\bar{k}$ $\hat{\pi}: \mathcal{U}=\operatorname{Spec}A\to \mathcal{U}/\!/G=\operatorname{Spec}A^G$
  1) $\hat{\pi}$ is surj. & $\forall$ G-inv. closed $Z\subset\mathcal{U}$, $\operatorname{im}Z\subset\mathcal{U}/\!/G$ is closed. It remains true after base change.
  2) G-inv $Z_1,Z_2\overset{closed}{\subset}\mathcal{U}$ $\operatorname{im}(Z_1\cap Z_2)=\operatorname{im}(Z_1)\cap\operatorname{im}(Z_2)$

     In particular $x_1, x_2\in\mathcal{U}(k)$ $\hat{\pi}(x_1)=\hat{\pi}(x_2)$ iff $\overline{Gx_1}\cap\overline{Gx_2}\neq\emptyset$.
  3) If A Noeth. so is A^G. If A is f.g./k then A^G is f.g./k and $\forall$ f.g A-mod. $M\circlearrowleft G$ M^G is f.g. A^G-mod
  4) if A noeth. then $\hat{\pi}$ is initial for G-inv maps to alg. spaces.

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§2. Coh. Affine Morphisms (condition 2))

- **Def**: A q-cpt q-sep morphism $f:\mathcal{X}\to\mathcal{Y}$ is coh. affine. if $f_*:QCoh(\mathcal{X})\to QCoh(\mathcal{Y})$

  A – – alg stack $\mathcal{X}$ is coh. affine if $\mathcal{X}\to\operatorname{Spec}\mathbb{Z}$ is – –
- Eg: · An affine morphism.
  - An affine alg gp G/k is linearly reductive iff $BG\to\operatorname{Spec}k$ is coh. affine (GMS)
- **Lem 6.3.15**:

$$\begin{tikzcd}
\mathcal{X}' \arrow[r, "f"] \arrow[d, "\pi'"'] \arrow[dr, phantom, "\ulcorner", very near start] & \mathcal{X} \arrow[d, "\pi"] \\
\mathcal{Y}' \arrow[r, "g"] & \mathcal{Y}
\end{tikzcd}$$

1) If g is f.f. π' is GMS ⇒ π is so
2) If $\mathcal{Y}$ has q-affine diag. π is GMS ⇒ π' is so.

Pf: 1) By flat base change $\pi'_*f^*\simeq g^*\pi_*$ ($\uparrow$ $\uparrow$ $\uparrow$ f. exact) π_* is exact.

2) First g is q-affine. $\mathcal{Y}'\underset{open}{\hookrightarrow}\operatorname{Spec}_{\mathcal{Y}}f_*\mathcal{O}_{\mathcal{Y}'}\xrightarrow{finite}\mathcal{Y}$
- g is affine. g_* f. exact. ✓
- g is open immersion. $F'\twoheadrightarrow G'$ in $QCoh(\mathcal{X}')$

$G:=\operatorname{im}(f_*F'\to f_*G')$ Since $f^*f_*\simeq id$ $f^*G\simeq G'$

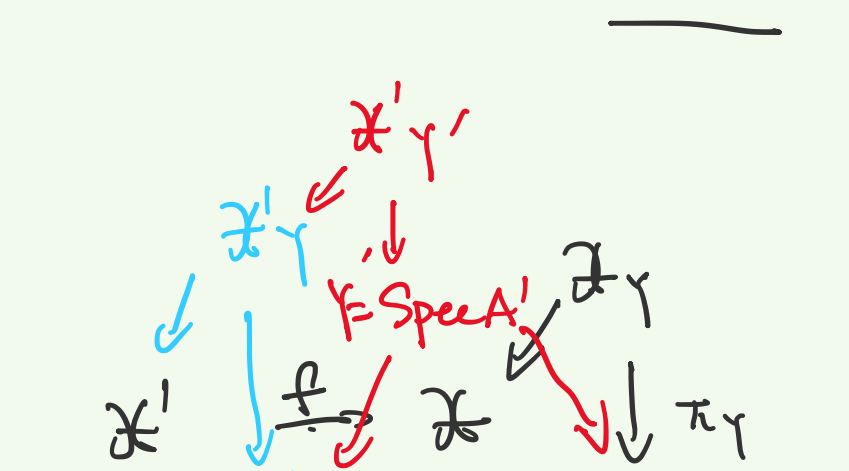
π_* is exact ⇒ $\pi_*f_*F'\twoheadrightarrow\pi_*G$

pull back via g ⇝

$$g^*\pi_*f_*F' \underset{\substack{\uparrow\\ \text{flat base change}}}{\simeq} \pi'_*f^*f_*F' \simeq \underline{\pi'_*F'} \twoheadrightarrow g^*\pi_*G \simeq \pi'_*f^*G\simeq\underline{\pi'_*G'}$$

To prove the general case

WMA $\mathcal{Y}'$ $\mathcal{Y}$ q-cpt and choose sm. rep. $Y=\operatorname{Spec}A\to\mathcal{Y}$ which is q-affine. π_Y is coh. affine.



$\mathcal{Y}'_Y\to\mathcal{Y}$ is f.f. by 1) STS $\mathcal{X}'_Y\to\mathcal{Y}'_Y$ is coh. affine

By 1) again Suffices to check after $Y'=\operatorname{Spec}A'\to\mathcal{Y}'_Y$

We win since $Y'\to Y$ is affine. ▨

- **Cor 6.3.16**: $f:\mathcal{X}\to\mathcal{Y}$ rep. $\mathcal{Y}$ has q-affine diag. f coh affine ⇒ affine. ▨

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§3. First properties of GMS

- **Lem 6.3.20**:

$$\begin{tikzcd}
\mathcal{X}' \arrow[r, "f"] \arrow[d, "\pi'"'] & \mathcal{X} \arrow[d, "\pi"] \\
X' \arrow[r, "g"'] & X
\end{tikzcd}$$

X' X q-sep. alg. spaces ($\Leftarrow X$ has q-affine diag)

1) If g is f.f. π' is GMS ⇒ π is GMS
2) π is GMS ⇒ π' is GMS

Assume π is GMS

3) (Projection formula) $F\in QCoh(\mathcal{X})$ $G\in QCoh(X)$ $\pi_*F\otimes G\xrightarrow{\sim}\pi_*(F\otimes\pi^*G)$

   In particular $G\xrightarrow{\sim}\pi_*\pi^*G$
4) $F\in QCoh(\mathcal{X})$ $g^*\pi_*F\xrightarrow{\sim}\pi'_*f^*F$
5) $\forall J\subset\mathcal{O}_X$ q-coh $J\xrightarrow{\sim}\pi_*(\pi^{-1}J\cdot\mathcal{O}_{\mathcal{X}})$

Pf: 1) g flat ⇒ $g^*(\mathcal{O}_X\overset{\uparrow}{\to}\pi_*\mathcal{O}_{\mathcal{X}})$ ⇝ $\mathcal{O}_{X'}\overset{\uparrow}{\to}\pi'_*\mathcal{O}_{\mathcal{X}'}$

$\mathcal{O}_X\xrightarrow{\sim}\pi_*\mathcal{O}_{\mathcal{X}}$ ⇔ $\mathcal{O}_{X'}\xrightarrow{\sim}\pi'_*\mathcal{O}_{\mathcal{X}'}$ condition 1)

By Lem 6.3.15 2) descends

- 2) holds when g is flat. ⇐

3) $U\xrightarrow{étale}X$ U disjoint union of affine schemes.

$\pi_U:\mathcal{X}_U\to U$ is GMS by flat case of 2).

$id\to\pi_*\pi^*$ ⇝ $id\to\pi_{U*}\pi_U^*$ WMA $X=\operatorname{Spec}A$

$G_2\to G_1\to G\to 0$ free pre.

$$\begin{tikzcd}
\pi_*F\otimes G_2 \arrow[r] \arrow[d, "\sim"] & \pi_*F\otimes G_1 \arrow[r] \arrow[d, "\sim"] & \pi_*F\otimes G \arrow[r] \arrow[d, "\sim"] & 0 \\
\pi_*(F\otimes\pi^*G_2) \arrow[r] & \pi_*(F\otimes\pi^*G_1) \arrow[r] & \pi_*(F\otimes\pi^*G) \arrow[r] & 0
\end{tikzcd}$$

⇐ π_* is exact.

2) By Lem 6.15 2) π'_* is exact. WTS $\mathcal{O}_{X'}\xrightarrow{\sim}\pi'_*\mathcal{O}_{\mathcal{X}'}$

WMA X, X' are affine g_* f. exact.

Apply 3) to $F=g_*\mathcal{O}_{X'}$

$$\cancel{g_*}\mathcal{O}_{X'}\xrightarrow{\sim}\pi_*\pi^*g_*\mathcal{O}_{X'}\underset{\substack{\uparrow\\ g\text{ affine.}}}{\simeq}\pi_*f_*\pi'^*\mathcal{O}_{X'}\simeq\pi_*f_*\mathcal{O}_{\mathcal{X}'}\simeq\cancel{g_*}\pi'_*\mathcal{O}_{\mathcal{X}'}$$

$\Rightarrow\mathcal{O}_{X'}\simeq\pi'_*\mathcal{O}_{\mathcal{X}'}$

4) is true if g is flat. moreover it is fppf local on X

WMA $X'=\operatorname{Spec}A'\xrightarrow{g}X=\operatorname{Spec}A$ factor g as a closed imm. + flat morphism. WMA $X'\overset{closed}{\hookrightarrow}X$ defined by $J\subset\mathcal{O}_X$.

WTS $\pi_*F/J\pi_*F\xrightarrow{\sim}\pi_*(F/(\pi^{-1}J)\mathcal{O}_{\mathcal{X}}F)$

$\pi^{-1}J\cdot\mathcal{O}_{\mathcal{X}}\cdot F=\operatorname{im}(\pi^*J\otimes F\to F)$

$\pi_*(\pi^{-1}J\cdot\mathcal{O}_{\mathcal{X}}\cdot F)=\operatorname{im}(\pi_*(\pi^*J\otimes F)\to\pi_*F)$
$\overset{3)}{=}\operatorname{im}(J\otimes\pi_*F\to\pi_*F)=J\pi_*F$

5) Standard diag. chasing. + π_* exact. ▨

- **Lem 6.3.22**: $\pi:\mathcal{X}\to X$ GMS X q-sep.
  1) $\mathcal{A}\in QCoh(\mathcal{O}_{\mathcal{X}})$ $\operatorname{Spec}_{\mathcal{X}}\mathcal{A}\xrightarrow{\pi_{\mathcal{A}}}\operatorname{Spec}_X\pi_*\mathcal{A}$ is GMS
  2) $\mathcal{Z}\overset{closed}{\subset}\mathcal{X}$ defined by $\mathcal{I}$ $Z=\operatorname{im}\mathcal{Z}\subset X$ $\mathcal{Z}\to Z$ is GMS.

Pf: $\mathcal{Z}=\operatorname{Spec}_{\mathcal{X}}(\mathcal{O}_{\mathcal{X}}/\mathcal{I})$ 2) ⇐ 1)

1) $\operatorname{Spec}_{\mathcal{X}}\mathcal{A}\underset{\substack{\uparrow\\ \text{affine}}}{\longrightarrow}\mathcal{X}\times_X\operatorname{Spec}_X\pi_*\mathcal{A}\underset{\substack{\uparrow\\ \text{coh. affine by Lem. 6.3.15 2)}}}{\longrightarrow}\operatorname{Spec}_X\pi_*\mathcal{A}$ ▨

- Pf of Thm 6.3.5 1) $\mathcal{X}$ q-sep ⇒ so is X.

  $\forall x\in X(k)$ use Lem 6.3.20 2) $\mathcal{X}\times_X\operatorname{Spec}k\to\operatorname{Spec}k$ GMS

  $\Gamma(\mathcal{X}_x,\mathcal{O}_{\mathcal{X}_x})=k$ ⇒ $\mathcal{X}_x\neq\emptyset$ ⇒ π is surj.

  $\mathcal{Z}\overset{closed}{\subset}\mathcal{X}$ Lem 6.3.22 2) $\mathcal{Z}\to\operatorname{im}\mathcal{Z}$ is GMS hence surj.

  So $\operatorname{im}\mathcal{Z}=\pi(\mathcal{Z})$ hence $\pi(\mathcal{Z})$ is closed. + base change.

  2) $\mathcal{Z}_i\subset\mathcal{X}$ defined by $\mathcal{I}_i$ $i=1.2$ apply π_*

  $0\to\mathcal{I}_1\to\mathcal{I}_1+\mathcal{I}_2\to\mathcal{I}_2/\mathcal{I}_1\cap\mathcal{I}_2\to 0$ $\mathcal{I}_2\twoheadrightarrow\mathcal{I}_2/\mathcal{I}_1\cap\mathcal{I}_2$

  $\pi_*\mathcal{I}_1+\pi_*\mathcal{I}_2\xrightarrow{\sim}\pi_*(\mathcal{I}_1+\mathcal{I}_2)$ i.e. $\operatorname{im}\mathcal{Z}_1\cap\operatorname{im}\mathcal{Z}_2=\operatorname{im}(\mathcal{Z}_1\cap\mathcal{Z}_2)$

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§3. Finite typeness of GMS

- **Def'n**: A morphism $f:X\to Y$ of schemes is univ. submersive if it is surj. & Y has the quotient top. ($U\subset Y$ open iff $f^{-1}U$ open)
- (Lem/Ex "Valuative Criterion")

$$\begin{tikzcd}
\operatorname{Spec}R' \arrow[r, dashed] \arrow[d] & X \arrow[d] \\
\operatorname{Spec}R \arrow[r] & Y
\end{tikzcd}$$

R'/R DVR

- Eg: GMS fppf covering.
- **Prop 6.3.23**:

$$\begin{tikzcd}
X \arrow[d, "f"'] \arrow[dr, "\pi"] & \\
Y \arrow[r] & S
\end{tikzcd}$$

$X. Y, S$ Noeth. f univ. submersive. $\pi:X\to S$ is of f.t. ⇒ $Y\to S$ of f.t.

Pf: WMA $S=\operatorname{Spec}R$ $Y=\operatorname{Spec}B$ B int. domain.

- By flatness & Raynaud-Gruson flatification.

$$\begin{tikzcd}
\tilde{X} \arrow[r] \arrow[d, "f'"'] & X \arrow[d, "f"] \arrow[dr, "\pi"] & \\
Bl_I Y=Y' \arrow[r, "g"] & Y \arrow[r] & S
\end{tikzcd}$$

$\tilde{X}\to Y'$ flat

- $\tilde{X}=(Y'\setminus g^{-1}V(I))\times_Y X\subset Y'\times_Y X$ of f.t. S
- Claim: $\tilde{X}\xrightarrow{f'}Y'$ is surj.

  (Use "Valuative Criterion")

$$\begin{tikzcd}
\tilde{X} \arrow[d, "\tilde{f}"'] \arrow[dr, "f.t"] & \\
Y' \arrow[r] & S
\end{tikzcd}$$

$\tilde{f}$ f.f. of f.t. (Noeth) $\overset{descent}{\Rightarrow}$ $Y'\to S$ is of f.t.

- Choose generators $f_1,\dots,f_n\in I$ s.t. $Y'=\bigcup\operatorname{Spec}B_i$

  $B_i=B\langle f_j/f_i\rangle_{j\neq i}\subset\operatorname{Frac}B.$

  Write $B=\varinjlim B_\lambda$ B_λ f.g. R-alg. $\lambda\gg0$ $f_i\in B_\lambda$

  $B_{\lambda,i}:=B_\lambda\langle f_j/f_i\rangle=B_i$ ⇒ $Y'=Bl_I\operatorname{Spec}B=Bl_{I_\lambda}\operatorname{Spec}B_\lambda$

  where $I_\lambda=(f_1,\dots,f_n)B_\lambda$

  $g_\lambda: Y'\xrightarrow{g}Y=\operatorname{Spec}B\xrightarrow{p_\lambda}\operatorname{Spec}B_\lambda$

  push forward $\mathcal{O}_Y\hookrightarrow g_*\mathcal{O}_{Y'}$ along p_λ

  ⇝ $p_{\lambda*}\mathcal{O}_Y\hookrightarrow\underline{g_{\lambda*}\mathcal{O}_{Y'}}$
  - Claim: $g_{\lambda*}\mathcal{O}_{Y'}$ coh. on $\operatorname{Spec}B_\lambda$ ⇒ $p_{\lambda*}\mathcal{O}_Y$ is coh

    ⇒ B/B_λ finite ⇒ B/R finite type. ▨

Pf of Thm 6.3.5 3) · If $\mathcal{X}$ Noeth. $I_1\subset I_2\subset\cdots$ on X

$\pi^{-1}I_1\cdot\mathcal{O}_{\mathcal{X}}\subset\pi^{-1}I_2\cdot\mathcal{O}_{\mathcal{X}}\subset\cdots$ terminates

By Lem 6.3.22 5) $I_n=\pi_*(\pi^{-1}I_n\cdot\mathcal{O}_{\mathcal{X}})$ ⇒ $I_1\subset I_2\subset\cdots$ terminates.

- Assume S Noeth. $\mathcal{X}$ of f.t./S

$$\begin{tikzcd}
U \arrow[d] \arrow[drr, bend left] \arrow[dd, bend right] & & \\
\mathcal{X} \arrow[d, "\substack{\text{univ}\\ \text{submersive}}"'] & & \\
X \arrow[rr] & & S
\end{tikzcd}$$

$X\to S$ is of f.t. by Prop 6.3.23.

- $F\in Coh(\mathcal{X})$ $\pi_*F\in Coh(X)$ WMA $X=\operatorname{Spec}A$ $\mathcal{X}$ irr.

① Assume $\mathcal{X}$ red. By Noeth. ind. WMA π_*F coh. if $\operatorname{Supp}F\neq\mathcal{X}$.

$0\to F_{tor}\to F\to F/F_{tor}\to 0$ STS $\pi_*(F/F_{tor})$ is coh.

i.e. WMA F torsion-free. ⇒ $\forall s:\mathcal{O}_{\mathcal{X}}\to F$ inj.

② Induction on $\dim\xi^*F$ $\xi:\operatorname{Spec}K\to\mathcal{X}$ image gen. pt.

$0\to\mathcal{O}_{\mathcal{X}}\to F\to\underline{F/\mathcal{O}_{\mathcal{X}}}\to 0$ π_*. + induction.

③ $I\subset\mathcal{O}_{\mathcal{X}}$ defining $\mathcal{X}_{red}$ $\exists N$ s.t. $I^N=0$

$0\to I^{k+1}F\to I^kF\to I^kF/I^{k+1}F\to 0$ + induction.

§5. Initiality

- Pf of Thm 6.3.5 4) WTS

$$\begin{tikzcd}
\mathcal{X} \arrow[d, "\pi"'] \arrow[dr, "f"] & \\
X \arrow[r, dashed, "\exists! h"'] & Y
\end{tikzcd}$$

Similar to the Pf Thm 4.2.6

- Uniqueness. $h_1. h_2$ two fillings

  $E\xrightarrow{e}X\overset{h_1}{\underset{h_2}{\rightrightarrows}}Y$ equalizer monomorphism loc. of f.t.

  $\pi:\mathcal{X}\to X$ factors through $e:E\to X.$

  e univ. closed. $\boxed{\mathcal{O}_X\hookrightarrow e_*\mathcal{O}_E}$

  A univ. closed. mono. loc. of f.t. is a closed immersion

  $e:E\xrightarrow{\sim}X$ ⇒ $h_1=h_2$
- Existence: · When Y is affine.

  $Mor(X,Y)=Hom(\Gamma(Y,\mathcal{O}_Y),\Gamma(X,\mathcal{O}_X))$
  $=Hom(\Gamma(Y,\mathcal{O}_Y),\Gamma(\mathcal{X},\mathcal{O}_{\mathcal{X}}))=Mor(\mathcal{X},Y)$
  - In general $\mathcal{X}$ q-cpt WMA Y q-cpt.

    Similar to Thm 4.2.6 WMA $X=\operatorname{Spec}A$ A strictly henselian local.

    $x\in\mathcal{X}$ unique closed pt by Thm 6.3.5 2)

    and let $\mathcal{G}_x\hookrightarrow\mathcal{X}$ closed immersion of residual gerbe (Prop 3.5.16)
  - $(Y'=\operatorname{Spec}B, y')\to(Y, f(x))$ étale pre.

$$\begin{tikzcd}
x'\in\mathcal{U}'\subset\mathcal{X}' \arrow[r, "f'"] \arrow[d] \arrow[dr, phantom, "\ulcorner", very near start] & Y' \arrow[dd] \\
\mathcal{X} \arrow[ur] \arrow[d, "\pi"'] \arrow[dr, "f"] & \\
X \arrow[uur, dashed] & Y
\end{tikzcd}$$

$\mathcal{X}'\to\mathcal{X}$ étale, surj. rep. sep.

$x'\in\mathcal{X}'$ a preimage of $x\in\mathcal{X}$

and $\mathcal{U}'\subset\mathcal{X}'$ q-cpt open nbhd of x'

- $\mathcal{U}'\to\mathcal{X}$ q-finite. sep. rep.

  so by Zariski MT $\mathcal{U}'\overset{open}{\hookrightarrow}\tilde{\mathcal{X}}\xrightarrow{finite}\mathcal{X}$ $\underline{\tilde{\mathcal{X}}=\operatorname{Spec}_{\mathcal{X}}\mathcal{A}}$

  By Lem 6.3.22 1) $\tilde{\mathcal{X}}\xrightarrow{\tilde{\pi}}\tilde{X}=\operatorname{Spec}_X\pi_*\mathcal{A}$ GMS

  and by Thm 6.3.5 2) $\pi_*\mathcal{A}$ is coh. ⇒ $\tilde{X}\to X$ finite

  $\tilde{X}=\coprod\operatorname{Spec}A_i$ each A_i henselian local.

  Replace $\tilde{X}$ by $\tilde{\mathcal{X}}:=\tilde{\pi}^{-1}\operatorname{Spec}A_i$ containing x'

  and $\mathcal{U}'$ by $\mathcal{U}'\cap\tilde{\mathcal{X}}$

  Then $\tilde{\mathcal{X}}$ has a unique closed pt which has to x'

  ⇒ $\tilde{\mathcal{X}}=\mathcal{U}'$ $\mathcal{U}'\to\mathcal{X}$ finite étale inducing an isom. of residual gerbes ⇒ $\mathcal{U}'\cong\mathcal{X}$

  $\mathcal{X}\to Y'$ factors through X because Y' is affine.

  So f factors – – X. ▨

- Applications

  1) Luna's Fundamental Lemma.
  - **Thm 6.3.24**:

$$\begin{tikzcd}
\mathcal{X}' \arrow[r, "f"] \arrow[d, "\pi'"'] & \mathcal{X} \arrow[d, "\pi"] \\
X' \arrow[r, "g"] & X
\end{tikzcd}$$

f sep rep.
$\mathcal{X}'$ $\mathcal{X}$ have affine diag
π' π GMS

$x'\in\mathcal{X}'$ s.t.

1) f étale at x'
2) f induces an isom. of stab. at x' and
3) $x'\in\mathcal{X}'$ and $f(x')=x\in\mathcal{X}$ are closed pts

Then $\exists$ $\pi'(x')\in U'\overset{open}{\subset}X'$ s.t. $U'\xrightarrow{g}X$ étale and $U'\times_X\mathcal{X}\cong\pi'^{-1}(U')$

2) Properties of linearly reductive gps